

Environmental Monitoring Dashboard (COIL)

A community environmental group has asked your team to create a web-based environmental monitoring system. They have sensor readings (including realistic sample/simulated data) and need a dashboard that helps people understand current conditions, spot issues, and respond appropriately.

System outline (core)

Your solution should produce a single integrated system with:

1. Community Dashboard (public-facing web app)
2. Data Management & Alerts (validation, storage, alert rules)

COIL workshops (Weeks 1–2 only) **

In Weeks 1–2, you will take part in facilitated Collaborative Online International Learning (COIL) workshops. These sessions will help your team:

- agree a specific environmental focus (e.g., flood risk, drought, soil quality, heat)
- identify stakeholders and plan requirements gathering
- finalise the scope, features, and delivery plan for your prototype

Community Dashboard (core)

A web interface for residents and community members.

It should allow people to:

- view current readings for your chosen environmental focus (e.g., water level, air quality, temperature, soil moisture)
- see a clear status (e.g., normal / warning / critical)
- view simple historical trends (basic charts over time)
- view readings by site/location (e.g., site list; map view if appropriate)
- read active alerts with a short explanation of what they mean

Key accessibility: clear text, good contrast, keyboard usability, mobile-friendly layout.

Data Management & Alerts (core)

This section receives/loads readings, checks them, stores them, and triggers alerts.

It should:

- validate incoming data (e.g., missing values, unexpected ranges, duplicates)
- store time-stamped readings for later retrieval
- keep basic sensor/site metadata (e.g., location, type, status)
- implement alert rules (thresholds + severity)
- maintain an alert log/history (what triggered, when, severity)

Storage approach: teams should justify their storage approach and show how it supports historical trends and alert review.

Extensions and improvements

To reach the highest marks, implement one or more integrated extensions.

Suggested main extension: *Analysis & Reporting Portal*

A separate view for researchers/decision-makers that builds on the same stored data and alert history, for example:

- filters (date range, location/site, sensor type)
- trend summaries and comparisons across sites
- export/download (e.g., CSV)

Other extension ideas (choose one or design your own):

- offline/low-bandwidth mode
- more advanced alert logic (e.g., rolling averages, persistence rules)
- partial multi-language support

**** COIL is Weeks 1–2 only**

During those two weeks you will take part in four facilitated online workshops with partner universities (CS students from China & South Africa). After Week 2, you continue as a normal COMP2850 project using your teams agreed scope and plan.

By the end of Week 2 your team should have:

- a chosen topic and clear stakeholder focus
- a short requirements summary (what matters, why, for whom)
- a first-pass design (sketches/wireframes + simple system diagram)
- an agreed collaboration plan (tools, roles, meeting rhythm, workload approach)
- an initial backlog and delivery plan for the remaining project weeks

This work sits within a funded research project on community-centred environmental monitoring and international collaboration. Where applicable, you will follow the project's agreed participation procedures (e.g., consent and recording during workshops), with opportunities for publication authorship and further participation.